António Ferreira

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EDUCATION

Northwestern University **Evanston, IL**
Ph.D., Materials Science & Engineering

Georgia Institute of Technology **Atlanta, GA**
B.S./M.S., Materials Science & Engineering, GPA: 3.95 (B.S.) / 3.88 (M.S.) May 2025 / May 2026

WORK EXPERIENCE

Engineering Intern – Havelar | *Porto, Portugal* *May 2026 – Sept. 2026*
- Printed a 4.5m tall, 43m^2 dome designed by Mario Cucinella Architects, sourcing and qualifying local materials from Monopoli, Italy
- Developed a new industrial-scale 3D concrete printer, including: printhead design increasing print speed by 2.5x, real-time admixture tuning via Delta DVP-SX2 PLC programming, and an operator-facing app for live machine control

Materials Engineer Intern – Framatome | *Lynchburg, VA, USA* *May 2025 – Aug. 2025*
- Automated irradiation embrittlement and fracture mechanic calculations (ART, USE, PTS, UCC) for Pressure-Temperature safety curves
- Evaluated corrosion of reactor materials under primary water conditions to support component integrity assessments

Materials Engineer Intern – COBOD | *Copenhagen, Denmark* *May 2024 – Aug. 2024*
- Led the development of a novel insulating material and its delivery system for 3D printing technologies
- Fabricated and tested cement-based materials, and 3D printed large-scale concrete structures

Research Intern – Institute of Computational Physics, ZHAW | *Winterthur, Switzerland* *May 2023 – Aug. 2023*
- Performed extensive optical and drift diffusion simulations of DBRs applied to Perovskite Solar Cells
- Measured the External Quantum Efficiency and J-V curves of Building Integrated Photovoltaics

RESEARCH EXPERIENCE

Researcher – Prof. Aaron Stebner Lab | *Atlanta, GA, USA* *Sept. 2023 – May 2026*
- Performing microstructural and chemical analysis of Wire-Arc Additive Manufactured (WAAM) NiTi SMAs
- CASMART 6th Student Design Challenge: fabricated elastocaloric NiTiCu SMAs, tested and characterized materials with DSC, SEM, cyclical compression, and presented research findings at SMST 2024
- Optimized ultrasonic atomization of metal powder, impurity detection and reduction, for feedstock on demand

Researcher – Prof. Natalie Stingelin Lab | *Atlanta, GA, USA* *Aug. 2021 – Sept. 2024*
- Fabricated and optimized polymer based thin films, distributed Bragg reflectors and optical microcavities
- Wrote MATLAB and Python programs for data analytics, optical simulations and modeling, and experiment design

LEADERSHIP EXPERIENCE

Teaching Assistant – Materials Selection and Design | *Atlanta, GA, USA* *Aug. 2025 – May 2026*
- Assist students in an advanced Material Selection and Design course using Ashby's method and Ansys software

Teaching Assistant – Chemical Thermodynamics of Materials | *Atlanta, GA, USA* *Aug. 2023 – May 2025*
- Assisted 125+ students in core MSE course; explained thermodynamic concepts and graded assignments

CTO – Materials Innovation and Learning Laboratory (MILL) | *Atlanta, GA, USA* *Sept. 2021 – Jan. 2024*
- Oversaw characterization team, enhanced staff/user experience, procured equipment, and managed the laboratory
- Trained students on optical and electron microscopy, optical profilometry, infrared spectroscopy, XRD, and XRF

AWARDS and PUBLICATIONS

- 2024 Outstanding Teaching Assistant for School of Materials Science and Engineering
- ThinkSwiss Research Scholarship, 2023
- 1st Place Undergraduate Research Poster, MRS (2022, Georgia Tech MSE)
- V. Quirós-Cordero, A. Balzer, S. Bachevillier, N. Watkins, **A. Ferreira**, J. Mushyakov, M. Dhoot, C. Silva-Acuña, P. Stavrinou, N. Stingelin. "A Versatile Materials Class for Solution-Processed Optics and Photonics Based on Titanium Oxide Hydrates and Polyalcohols: A Perspective." *Advanced Materials* (2025), doi.org/10.1002/adma.202507028

SKILLS

Computational & Modeling: Python, MATLAB, DFT (Quantum Espresso), CALPHAD (Thermo-Calc), Machine Learning (CNN, U-Net), CAD (Solidworks, NX), Node-Red, Excel

Characterization & Processing: SEM, DSC, XRD, XRF, FTIR, UV-Vis, Profilometry, Metallography, Vacuum Arc Melting, Wire EDM, PLC Programming (Delta DVP-SX2, ISPSoft), Modbus RTU